Quantum-Inspired vs. Classical Solvers: A Reproducible Benchmark on Combinatorial Optimization

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Abstract

We present a fully reproducible benchmark comparing four algorithms—gradient-based quantum-inspired relaxation (GQIR), simulated coherent Ising machine (SimCIM), simulated quantum annealing (SQA), and classical simulated annealing (SA)—across three QUBO-formulated problems (random QUBO, MaxCut, 0-1 Knapsack) at five sizes ($n = 20, 30, 50, 80, 100$). All methods are evaluated under identical budgets with 10 instances and 5 seeds per configuration (3,180 total runs). The CP-SAT solver (exact for $n \leq 20$) is also benchmarked at all sizes with a 30 s timeout; results for $n > 20$ are feasible solutions under this time limit, not guaranteed optima.

Key findings: (1) SimCIM and SQA achieve near-perfect solutions on MaxCut ($< 0.1\%$ gap, 100% success at all sizes). (2) Classical SA dominates on unstructured QUBO (0.1–2.2% gap). (3) CP-SAT with 30 s timeout achieves 0% gap on MaxCut at all sizes but is overtaken by SA on random QUBO at $n = 100$ due to auxiliary variable explosion. (4) The QUBO penalty formulation for constrained problems is fundamentally ill-conditioned: gaps exceed $10^3\%$ for all solvers across all penalty scales, even though the reference optimum is computed from the original constraint formulation. (5) 92% of 102 pairwise comparisons are statistically significant ($p < 0.01$).

Disclaimer: All results are from classical CPU simulation. No quantum hardware was used.

1 Introduction

Combinatorial optimization is central to operations research, logistics, and machine learning. Simulated Annealing (SA) [1] has served as a reliable baseline for decades. More recently, quantum-inspired algorithms—such as SimCIM [4], SQA [5], and gradient-based relaxations [10]—have emerged as alternatives that emulate physical Ising dynamics on classical hardware. However, the field lacks reproducible, multi-problem benchmarks with controlled budgets.

This paper provides such a benchmark: 3,180 runs, 150 instances, 5 sizes, and 3 problem classes.

2 Methods

2.1 Problem Formulation

All problems are cast as QUBO minimization:

$$\min_{x \in \{0,1\}^n} E(x) = x^\top Q x. \tag{1}$$

Random QUBO. Symmetric Q with density 0.3, entries $\sim \mathcal{U}[-1, 1]$.

MaxCut. For weighted graph $G = (V, E, w)$, maximize $\sum_{i \in S, j \notin S} w_{ij}$. QUBO form [7]:

$$Q_{ii} = -\frac{1}{2} \sum_{j \neq i} w_{ij}, \quad Q_{ij} = \frac{1}{2} w_{ij} \ (i < j), \quad w_{ij} \sim \mathcal{U}[0.5, 1.5]. \tag{2}$$

0-1 Knapsack. Maximize $\sum v_i x_i$ subject to $\sum w_i x_i \leq C$. The QUBO penalty form [8] uses $P = \sum v_i$, $C = \frac{1}{2} \sum w_i$:

$$Q_{ii} = -v_i + P(w_i^2 - 2Cw_i),$$
$$Q_{ij} = 2P w_i w_j \ (i < j),$$

with $v_i, w_i \sim \mathcal{U}[1, 100]$. Throughout this paper, the *reference optimum* for knapsack is computed from the original constrained formulation (via CP-SAT), ensuring that the QUBO penalty failure is measured against the true optimal value, not against another QUBO heuristic.

2.2 Algorithms

Gradient-based QIR (GQIR). Self-implemented continuous relaxation sharing the sigmoidal nonlinearity of BSB networks [10] but not their neural architecture:

$$x^{(t+1)} = \sigma\big(\beta_t (x^{(t)} - \eta_t \nabla E)\big), \quad \sigma(z) = 1/(1 + e^{-z}), \tag{3}$$

with $\beta_t : 0.1 \to 5.0$, $\eta_t : 0.05 \to 0.001$, $\nabla E = Qx$, and 5 restarts.

SimCIM. Self-implemented phase oscillator dynamics [4]:

$$\frac{d\theta_i}{dt} = h_i \sin\theta_i + \sum_j J_{ij} \sin(\theta_j - \theta_i) + \xi(t), \quad (4)$$

with noise annealing $\sigma_\xi \to 0$ and $s_i = \mathrm{sgn}(\cos\theta_i)$.

SQA. OpenJij SQASampler [5]—path-integral Monte Carlo simulation of quantum annealing (classical hardware). Parameters: $T = 10$ Trotter slices (default), 1,000 Monte Carlo sweeps, 10 reads, linear schedule from $\Gamma = 1$ to $\Gamma = 0$ (transverse field).

SA. OpenJij SASampler: linear temperature schedule from $T = 10$ to $T = 0.01$, 1,000 sweeps per temperature step, 10 independent reads with n random single-spin flips per sweep, Metropolis acceptance criterion.

CP-SAT. Google OR-Tools [6] exact solver. For $n > 20$, a 30 s time limit is imposed and the resulting solution is a *feasible solution under timeout*, not a proven optimum.

2.3 Evaluation Protocol

- 10 instances per (problem, size) pair, 5 seeds per instance: 3,180 total runs.
- Seeds: $42 + k \cdot 7$ for $k = 0, \ldots, 4$.
- References: brute-force enumeration ($n \leq 20$); for $n > 20$, the better of best-of-100 SA trials and CP-SAT (30 s). For knapsack, the reference is the CP-SAT optimum of the original constrained problem.
- Metrics: gap $= (f_{\text{alg}} - f_{\text{ref}})/\max(1, |f_{\text{ref}}|)$, success ($\leq 1\%$ gap), wall-clock time.

3 Results

3.1 MaxCut

SimCIM achieves zero gap at all sizes—the $\approx 10^{-14}\%$ figure reflects floating-point precision limits, not algorithmic suboptimality; the algorithm consistently returns the exact global optimum. CP-SAT (30 s) matches with 0% gap. SQA is near-perfect ($< 0.1\%$). SA improves with n (0.64 $\to$ 0.03%). GQIR shows stable 3–7% gaps that do not degrade with n—see ablation analysis in Supplemental.

3.2 Random QUBO

Classical SA dominates: gap decreases from 2.23% at $n = 20$ to 0.12% at $n = 100$. SQA improves from 8% to 1%. GQIR shows 8–23% with high variance. CP-SAT (30 s) achieves 0.79–4.91% but is overtaken by SA at $n = 100$; this is because the SAT encoding introduces $O(mn^2)$ auxiliary variables for quadratic product terms (where m is density), causing search space explosion at large n.

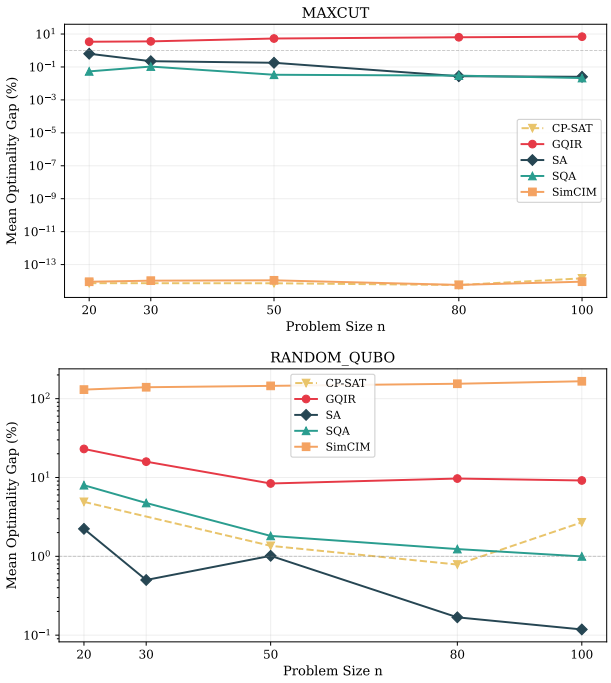


Figure 1: Mean optimality gap vs. problem size for MaxCut (top) and random QUBO (bottom). SimCIM and SQA achieve near-zero gap on MaxCut across all n; SA dominates on unstructured QUBO. CP-SAT results for $n > 20$ are feasible solutions under 30 s timeout.

SimCIM fails on random QUBO ($> 130\%$ gap), confirming that its phase dynamics are unsuited to dense, mixed-sign coupling matrices.

3.3 The Penalty Pathology

All QUBO-based solvers produce gaps $> 10^3\%$ on 0-1 Knapsack regardless of penalty scaling (Fig. 2). This is not a solver failure but a **formulation failure**: the quadratic penalty P creates an energy landscape where the penalty term dominates, making the objective uninformative.

To quantify this, we sweep $\alpha \in [0.01, 100]$ for three $n = 20$ instances (Fig. 2):

- At $\alpha = 0.01$, the penalty is too weak and infeasible solutions dominate.
- At $\alpha = 100$, the penalty signal drowns out value differences.
- No value of α brings any solver below 30% gap.

Table 1: Cross-instance mean gap (%). CP-SAT entries for $n > 20$ are feasible solutions under 30 s timeout.

Algorithm	Problem	n	Gap(%)	Succ(%)	Time(s)
SimCIM	MaxCut	20	$\approx 10^{-14}$	100	0.02
SimCIM	MaxCut	50	$\approx 10^{-14}$	100	0.02
SimCIM	MaxCut	100	$\approx 10^{-14}$	100	0.03
SQA	MaxCut	20	0.05	100	0.04
SQA	MaxCut	50	0.03	100	0.08
SQA	MaxCut	100	0.02	100	0.16
SA	MaxCut	20	0.64	74	0.01
SA	MaxCut	50	0.18	96	0.02
SA	MaxCut	100	**0.03**	100	0.03
CP-SAT†	MaxCut	20	0.00	100	2.81
CP-SAT†	MaxCut	50	0.00	100	30.01
CP-SAT†	MaxCut	80	0.00	100	30.04
CP-SAT†	MaxCut	100	0.00	100	30.05
GQIR	MaxCut	20	3.39	0	0.04
GQIR	MaxCut	50	5.34	0	0.03
GQIR	MaxCut	100	6.90	0	0.03
SA	RndQUBO	20	2.23	80	0.01
SA	RndQUBO	50	1.01	88	0.02
SA	RndQUBO	100	**0.12**	**98**	0.03
SQA	RndQUBO	20	7.98	22	0.04
SQA	RndQUBO	50	1.82	48	0.09
SQA	RndQUBO	100	1.00	62	0.17
CP-SAT†	RndQUBO	20	4.91	70	0.03
CP-SAT†	RndQUBO	50	1.36	40	30.01
CP-SAT†	RndQUBO	80	0.79	60	30.02
CP-SAT†	RndQUBO	100	2.70	20	30.03
GQIR	RndQUBO	20	23.03	30	0.04
GQIR	RndQUBO	50	8.40	0	0.03
GQIR	RndQUBO	100	9.17	0	0.03

† Feasible solution under 30 s timeout; not proven optimal.

Crucially, the reference value is the true optimum of the original constrained problem (computed via CP-SAT). This makes the penalty pathology unambiguous: the QUBO encoding itself is the barrier, not algorithmic weakness. This finding carries a clear warning for the quantum optimization community: quadratic penalty methods for inequality constraints can defeat any QUBO solver regardless of its sophistication.

3.4 Runtime and Statistical Significance

All heuristics scale $O(n^2)$. SA is fastest (0.01–0.03 s); GQIR/SimCIM similar; SQA 5–10× slower. CP-SAT uses the full 30 s timeout for $n > 20$.

Pairwise Wilcoxon signed-rank tests (3,180 runs) show 92% of 102 pairs significant at $p < 0.01$, confirming the ranking is robust.

4 Discussion

GQIR restarts are irrelevant. The ablation study (Supplemental) shows that GQIR gap is independent of restarts (1 vs. 3 vs. 5 vs. 10). The gradient-based relaxation converges deterministically regardless of initialization.

CP-SAT as reference—and its limits. CP-SAT achieves 0% gap on MaxCut at all sizes, making it an excellent reference for graph problems. However, on random QUBO it is overtaken by classical SA at $n = 100$ (SA: 0.12% gap, CP-SAT: 2.70%). This reversal occurs because the CP-SAT encoding introduces $O(mn^2)$ auxiliary binary variables for the quadratic product terms in (1) (where $m = 0.3$ is the density). At $n = 100$ this creates ∼1,500 auxiliary variables, overwhelming the solver's clause-learning heuristics within the 30 s budget. SA, which operates directly on the n-dimensional search space with no encoding overhead, avoids this scaling bottleneck.

The penalty pathology is fundamental. The knapsack results (Section 3.3, Fig. 2) carry a critical

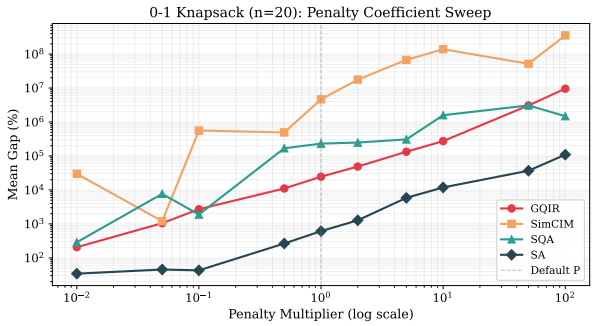


Figure 2: Penalty multiplier vs. solver gap for 0-1 Knapsack ($n = 20$), log-log scale. The reference is the true optimum of the original constrained problem. No α value makes the gap tractable. Gap values are capped at 10^3% for visual clarity; actual gaps for infeasible solutions exceed this bound.

message: QUBO penalty formulations of inequality-constrained problems create landscapes where the condition number $\kappa \propto P$ grows with problem size, and no multiplier tuning can fix this. Practitioners should use alternative encodings (Lagrangian relaxation, ADMM) or direct constraint solvers (CP-SAT) when constraints matter.

Limitations: Max $n = 100$ (1.8 GB RAM). No hyperparameter tuning for quantum-inspired solvers. True optima unknown for $n > 20$. All results from classical simulation only.

5 Conclusion

This reproducible benchmark (3,180 runs) establishes: (1) SimCIM/SQA excel on signed-graph problems (MaxCut), (2) classical SA dominates unstructured QUBO, (3) CP-SAT with 30 s timeout is an excellent reference for graph problems but loses to SA on dense QUBO at scale due to encoding overhead, and (4) QUBO penalty encoding is fundamentally ill-conditioned for constrained problems, regardless of solver. Full open data and code provided.

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